

A Quick Look at Open Source Numerical Computation for Windows

In this article, we present some numerical and scientific computational tools that the open source world has to offer to Windows users.



Mathematics and scientific computation have become part of every engineering workflow. The extensive numerical processing required in many fields like machine learning and analytics is driving technology trends. Windows users needn't have to worry, since there are many free and open source software when it comes to numerical or scientific computations. This article presents the various options that Windows users could try exploring in this context.

Scilab

Scilab is an open source numeric computational package governed by the CeCILL licence (GPL compatible). It is a great alternative to MATLAB and is available for use on Windows, Linux and Mac OS. The software provides a rich set of functionalities in signal processing, statistical analysis, fluid dynamics simulation, image enhancement and other frequently used computations in the automotive, aerospace, medical, control systems and meteorology industries. Scilab is also popular in academics for linear algebra, trigonometry, calculus, differentiation, genetic algorithms and statistics. It is currently gaining momentum in research fields like machine learning for its regression,

classification and probability distribution capabilities.

Apart from being software that can be accessed through a MATLAB-like GUI or an interactive command line, Scilab is also ideal as a platform for applications because of its computational abilities. It is interoperable with programming languages like C, C++, Java, Python and Tk1, and corresponding Scilab APIs can be called from these applications. Proprietary software like NI's LabVIEW can also be made to call into Scilab to make the best use of both the packages. Scilab also comes with a module called Xcos, which is a graphical simulation environment similar to MATLAB's Simulink. The software is known for its compatibility with MATLAB as it comes with a built-in code translator that smoothly migrates MATLAB code to versions compatible with Scilab.

The software can be downloaded from <http://www.scilab.org/download/latest>.

GNU Octave

Another open source option for Windows users is GNU Octave, which is freely available and redistributable software under the terms of the GPL as published by the Free